Multiple Myeloma — Cohen et al., 2021

GO / NO-GO GATES

Gate 1 — sample mixing & manifold continuity — GO

54 samples intermix per-neighbourhood (iLISI 1.0→14.1 of 54) on one continuous connected component; single Leiden lobe, aspect 1.09, only 2.8% cycling — no islands, no batch archipelago, no cycling compartment.

Gate 2 — embryonic module geography — GO

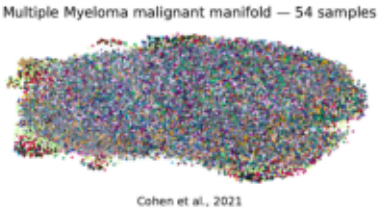
Non-branching: a single Gast/pSom-origin → terminal-Crest axis. Crest inversely opposes Gast (r=−0.69) and pSom (r=−0.61); pSom&Crest joint-high enrichment 0.22 (<1 = inverse gradient, not a co-expression hub).

Gate 3 — hub Gast↔pSom cross-repression — NA

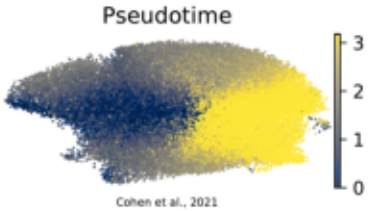
No bifurcation: Gast↔pSom near-orthogonal (r=+0.07), no competing arms — branching-only gate not applicable.

STEP FIGURES

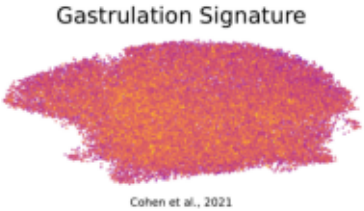
Gate 1 — sample integration



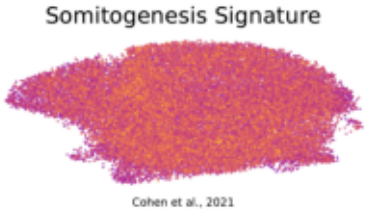
Trajectory & pseudotime (root at origin)



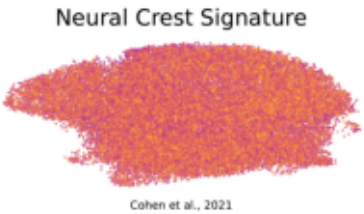
Gastrulation signature



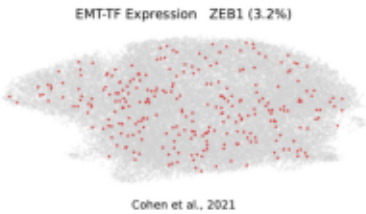
Somitogenesis signature



Neural-crest signature



EMT-TF expression (ZEB1)



Dataset. Multiple myeloma bone-marrow plasma cells (MARS-seq); 33,076 malignant cells, 54 samples ≥100 cells.

Verdict. The embryonic pattern reproduces as a NON-BRANCHING axis: Gast/pSom-high origin → terminal neural-crest, no resolved bifurcation. Linear-pSom/Crest geometry (Gast μ=−0.022, pSom μ=+0.014, Crest μ=+0.001).

Parameter choices

Sample threshold	≥100 cells (floor; smooth count decline, no dominant spillover cliff) → 54/73 samples, 33,147 cells
Overdispersed genes	482 (scFates find_overdispersed) • island removal: 71 cells / 2 detached specks → 33,076
Integration	Harmony θ=10 on sample (swept 2/5/10; strongest mixing, same-sample nbr frac 0.91→0.11)
Embedding	n_pcs=20 (scree plateau), n_neighbors=30, UMAP spread=1.3
Trajectory	scFates curve (ElPiGraph, Nodes=20, λ=0.3, μ=100) in Harmony-PCA space — non-branching, fold-free (2.5)
Root	curve tip with highest Gast+pSom, lowest Crest (developmental origin); Gast↓ ρ=−0.55, Crest↑ ρ=+0.57
EMT-TF readout	ZEB1 (3.2% expressing) — sole factor above detection floor; Snail/Twist/FOXC/PRRX/ZEB2 near-silent (<0.3%)
Annotation	Malignant plasma cells (Cohen et al. 2021 author cell_type); single differentiation continuum